

操作系统第6章作业

1：老和尚-小和尚挑水

信号量：

```
1 | int mutex1=1, mutex2=1, count=3, empty=10, full=0;
```

取水：

```
1 | P(full);  
2 | P(count);  
3 | P(mutex2);  
4 | V(mutex2);  
5 | V(count);  
6 | V(empty);
```

向缸中倒水：

```
1 | P(empty);  
2 | P(count);  
3 | P(mutex1);  
4 | V(mutex1);  
5 | P(mutex2);  
6 | V(mutex2);  
7 | V(full);  
8 | V(count);
```

2：多类型读-写者问题

初始：

```
1 | int mutex = 1;  
2 | int A_mutex = 1;  
3 | int B_mutex = 1;  
4 | int C_mutex = 1;  
5 | int data_mutex = 1;  
6 |  
7 | int countA = 0;  
8 | int countB = 0;  
9 | int countC = 0;
```

A类用户：

```
1 | P(A_mutex);
```

```

2  P(mutex);
3  countA++;
4  if (countA == 1) {
5      P(data_mutex);
6  }
7  V(mutex);
8  V(A_mutex);
9
10 read_data();
11
12 P(mutex);
13 countA--;
14 if (countA == 0) {
15     V(data_mutex);
16 }
17 V(mutex);

```

B类用户:

```

1  P(B_mutex);
2  P(mutex);
3  countB++;
4  if (countB == 1) {
5      P(data_mutex);
6  }
7  V(mutex);
8  V(B_mutex);
9
10 read_data();
11
12 P(mutex);
13 countB--;
14 if (countB == 0) {
15     V(data_mutex);
16 }
17 V(mutex);

```

C类用户:

```

1  P(C_mutex);
2  P(data_mutex);
3
4  write_data();
5
6  V(data_mutex);
7  V(C_mutex);

```

3: 牙科门诊

初始:

```
1 doctor=3, patient=0, mutex1=1,mutex2=1,mutexn=1;
2 int DState[3] = {0};
3 int PState[20] = {0};
4 int n = 0;
```

医生:

```
1 int i;
2 P(patient);
3 P(doctor);
4 P(mutex1);
5 DState[i] = 1;
6 V(mutex1);
7
8 work();
9
10 P(mutex1);
11 DState[i] = 0;
12 V(mutex1);
```

患者:

```
1 if (n < 20) {
2     int i = n;
3     P(mutexn);
4     n++;
5     V(mutexn);
6     V(patient);
7     P(mutex2);
8     PState[i] = 1;
9     V(mutex2);
10
11     look();
12
13     V(doctor);
14     P(mutex2);
15     PState[i] = 0;
16     V(mutex2);
17 }
```